

ATM Machine Project Report

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This is a simple command line - Python project showing the working of a simple ATM machine simulation. After card insertion, one can check his/her account balance, deposit and withdraw money in one terminal.

Features

Feature	Description
PIN check	3 attempts allowed.
Checking Balance	Shows the current account balance.
Depositing	Adds money to the account.
Withdrawing	Rs. 5000 max per withdrawal.
Daily limit	Rs. 25000 max per day, tracked automatically.
Exit	Exits the option and "ejects" the card.

Requirements

Python 3.x (no external libraries needed)

Setup & Run

Install Python 3.x. Clone or download this repo. In the project folder, run:

```
python atm.py
```

Usage

Press Enter to insert the card, then enter the PIN (3 tries max). Once logged in, pick an option from the menu and follow the prompts. Choose Exit to quit.

Test Login

Name: Ali Nawaz

Account: XXXX1234

PIN: 1234

Algorithm

1. Start the program.
2. Press Enter to insert the card.
3. Enter the PIN. Allow a maximum of 3 attempts.
4. If the PIN is correct, show the ATM menu.
5. Select balance, deposit, withdraw, daily limit, or exit.
6. For withdrawal, check the withdrawal amount, daily limit, and account balance.
7. Update the balance after a valid deposit or withdrawal.
8. Return to the menu after completing an operation.
9. Choose Exit to eject the card and end the program.

Pseudocode

BEGIN

SET name, card number, PIN, balance
SET amount withdrawn today = 0
PRESS ENTER TO INSERT CARD

WHILE PIN attempts left
 INPUT PIN
 IF PIN is correct
 LOGIN
 BREAK
 ELSE
 DISPLAY "Wrong PIN"
 END IF
END WHILE

IF login is not successful
 DISPLAY "Too many wrong attempts"
 END
END IF

WHILE TRUE
 DISPLAY MENU
 INPUT choice

 IF choice = BALANCE
 DISPLAY balance

 ELSE IF choice = DEPOSIT
 INPUT amount
 IF amount > 0
 ADD amount to balance
 ELSE
 DISPLAY "Invalid amount"
 END IF

 ELSE IF choice = WITHDRAW
 INPUT amount
 IF amount is valid (positive, multiple of 100)
 AND amount <= 5000
 AND withdrawn today + amount <= 25000
 AND amount <= balance
 SUBTRACT amount from balance
 ADD amount to withdrawn today
 DISPLAY "Collect your cash"
 ELSE
 DISPLAY "Transaction not allowed"
 END IF

 ELSE IF choice = DAILY LIMIT
 DISPLAY withdrawn today and amount left

 ELSE IF choice = EXIT
 DISPLAY "Card ejected"
 BREAK
END WHILE

```
        END IF
    END WHILE

END
```

Project Information

This project is made for a simple ATM machine simulation. It uses preset values for the account number, PIN and account balance. The program runs in one terminal and demonstrates the basic options available after card insertion and PIN verification.